Q.1) Write HTML script to display the text in bold, italic, underline and with strike. Apply separate effect on different text.

```html
<!DOCTYPE html>
<html>
<head>
   <title>Text Formatting</title>
</head>
<body>

   <h2>Text Formatting Example</h2>

   <p><b>This text is in Bold</b></p>

   <p><i>This text is in Italic</i></p>

   <p><u>This text is Underlined</u></p>

   <p><strike>This text has Strike-through</strike></p>

</body>
</html>
```

Q.2) Write a menu driven program to perform the following operations on associative arrays:

a) Merge the given arrays.

b) Find the intersection of two arrays.

c) Find the union of two arrays.

d) Find set difference of two arrays.

```php
<?php

function mergeArrays($arr1, $arr2) {
   return array_merge($arr1, $arr2);
}

function intersectArrays($arr1, $arr2) {
   return array_intersect_assoc($arr1, $arr2);
}

function unionArrays($arr1, $arr2) {
```

```
    return $arr1 + $arr2; // union
}

function differenceArrays($arr1, $arr2) {
    return array_diff_assoc($arr1, $arr2);
}

// Sample associative arrays
$array1 = array("a"=>1, "b"=>2, "c"=>3);
$array2 = array("b"=>2, "c"=>4, "d"=>5);

echo "Menu:\n";
echo "1. Merge Arrays\n";
echo "2. Intersection\n";
echo "3. Union\n";
echo "4. Difference\n";

echo "Enter your choice: ";
$choice = trim(fgets(STDIN));

switch($choice) {
    case 1:
        print_r(mergeArrays($array1, $array2));
        break;

    case 2:
        print_r(intersectArrays($array1, $array2));
        break;

    case 3:
        print_r(unionArrays($array1, $array2));
        break;

    case 4:
        print_r(differenceArrays($array1, $array2));
        break;

    default:
        echo "Invalid choice!";
}
```

```
?>
```

Q.1) Create an html page with following specifications :

a. Title should be about “My City”

b. Place your City name at the top of the page in large text and in red color

c. Add names and images (as a link) of landmarks in your city each in a different color, style and typeface .

d. After clicking on images it should display history of that place.

```html
<!DOCTYPE html>
<html>
<head>
  <title>My City</title>
</head>
<body>

  <!-- City Name -->
  <h1 style="color:red; font-size:50px; text-align:center;">
    Nashik
  </h1>

  <!-- Landmark 1 -->
  <h2 style="color:blue; font-family:Arial;">
    Panchavati
  </h2>
  <a href="panchavati.html">
    <img src="panchavati.jpg" alt="Panchavati" width="300">
  </a>

  <!-- Landmark 2 -->
  <h2 style="color:green; font-family:Courier;">
    Sula Vineyards
  </h2>
  <a href="sula.html">
    <img src="sula.jpg" alt="Sula Vineyards" width="300">
  </a>

  <!-- Landmark 3 -->
  <h2 style="color:purple; font-family:Georgia;">
    Trimbakeshwar Temple
  </h2>
  <a href="trimbak.html">
```

```
        <img src="trimbak.jpg" alt="Trimbakeshwar Temple" width="300">
    </a>

</body>
</html>
```

QQ1Write a PHP program to create a class temperature which contains data members as   Celsius and Fahrenheit . Create and Initialize all values of temperature object by using parameterized constructor . Convert Celsius to Fahrenheit and Convert Fahrenheit to Celsius using member functions. Display conversion on next page.

```
<?php

class Temperature {
    public $celsius;
    public $fahrenheit;

    // Parameterized constructor
    function __construct($c, $f) {
        $this->celsius = $c;
        $this->fahrenheit = $f;
    }

    // Convert Celsius to Fahrenheit
    function cToF() {
        return ($this->celsius * 9/5) + 32;
    }

    // Convert Fahrenheit to Celsius
    function fToC() {
        return ($this->fahrenheit - 32) * 5/9;
    }
}

// Get values from form
$c = $_POST['celsius'];
$f = $_POST['fahrenheit'];

// Create object using parameterized constructor
$temp = new Temperature($c, $f);
```

```
?>

<!DOCTYPE html>
<html>
<head>
  <title>Conversion Result</title>
</head>
<body>

<h2>Temperature Conversion Result</h2>

<?php
echo "Celsius to Fahrenheit: " . $temp->cToF() . " °F<br>";
echo "Fahrenheit to Celsius: " . $temp->fToC() . " °C";
?>

</body>
</html>
```

Q.1) Write HTML code to display following output. (15)

- **Tea**
    - **Hot tea**
    - **Black tea**
- **Coffee**
    - **Cold coffee**
    - **Hot coffee**

```
<!DOCTYPE html>
<html>
<head>
  <title>Tea and Coffee List</title>
</head>
<body>

<h2>Beverages</h2>

<ul type="square">
  <li>Tea
    <ul type="circle">
```

```
            <li>Hot tea</li>
            <li>Black tea</li>
        </ul>
    </li>

    <li>Coffee
        <ul type="disc">
            <li>Cold coffee</li>
            <li>Hot coffee</li>
        </ul>
    </li>
</ul>

</body>
</html>
```

Declare array. Reverse the order of elements, making the first element last and last element first and similarly rearranging other array elements. [Hint : array_reverse()]

```
<?php

// Declare an array
$arr = array(10, 20, 30, 40, 50);

// Display original array
echo "Original Array: <br>";
print_r($arr);

// Reverse the array
$reversed = array_reverse($arr);

// Display reversed array
echo "<br><br>Reversed Array: <br>";
print_r($reversed);

?>
```

Q.1) Write HTML code to display the list of different courses available in our college using ordered as well as unordered list.

```
<!DOCTYPE html>
<html>
```

```html
<head>
  <title>Courses in Our College</title>
</head>
<body>

<h2>Courses Available in Our College</h2>

<ol>
  <li>Undergraduate Courses
    <ul>
      <li>B.Sc (Computer Science)</li>
      <li>B.Com</li>
      <li>B.A</li>
    </ul>
  </li>

  <li>Postgraduate Courses
    <ul>
      <li>M.Sc (Computer Science)</li>
      <li>M.Com</li>
      <li>M.A</li>
    </ul>
  </li>

  <li>Diploma Courses
    <ul>
      <li>Diploma in Web Development</li>
      <li>Diploma in Banking</li>
    </ul>
  </li>
</ol>

</body>
</html>
```

Q.2) Write PHP script to demonstrate the concept of introspection for examining object.

```php
<?php

class Student {
```

```
    public $name = "Rahul";
    public $age = 20;

    function display() {
        echo "Student Name: " . $this->name;
    }

    function getAge() {
        return $this->age;
    }
}

// Create object
$obj = new Student();

// Introspection functions
echo "<h3>Class Name:</h3>";
echo get_class($obj);

echo "<h3>Class Methods:</h3>";
print_r(get_class_methods($obj));

echo "<h3>Class Properties:</h3>";
print_r(get_class_vars(get_class($obj)));

echo "<h3>Object Variables:</h3>";
print_r(get_object_vars($obj));

?>
```

Q.1) Design a admission form. which should contains : text box, multiline text box, a table which shows your academic record, radio button, check box, submit button.

```
<!DOCTYPE html>
<html>
<head>
    <title>Admission Form</title>
</head>
<body>
```

```html
<h2>College Admission Form</h2>

<form>

   <!-- Text Box -->
   Name: <input type="text" name="name"><br><br>

   <!-- Radio Buttons -->
   Gender:
   <input type="radio" name="gender" value="Male"> Male
   <input type="radio" name="gender" value="Female"> Female
   <br><br>

   <!-- Checkboxes -->
   Hobbies:
   <input type="checkbox" name="hobby[]" value="Reading"> Reading
   <input type="checkbox" name="hobby[]" value="Sports"> Sports
   <input type="checkbox" name="hobby[]" value="Music"> Music
   <br><br>

   <!-- Multiline Text Box -->
   Address:<br>
   <textarea name="address" rows="4" cols="40"></textarea>
   <br><br>

   <!-- Academic Record Table -->
   <h3>Academic Record</h3>
   <table border="1">
      <tr>
         <th>Exam</th>
         <th>Board/University</th>
         <th>Percentage</th>
      </tr>
      <tr>
         <td>10th</td>
         <td><input type="text" name="board10"></td>
         <td><input type="text" name="per10"></td>
      </tr>
      <tr>
```

```
        <td>12th</td>
        <td><input type="text" name="board12"></td>
        <td><input type="text" name="per12"></td>
      </tr>
      <tr>
        <td>Graduation</td>
        <td><input type="text" name="boardgrad"></td>
        <td><input type="text" name="pergrad"></td>
      </tr>
    </table>

    <br>

    <!-- Submit Button -->
    <input type="submit" value="Submit">

</form>

</body>
</html>
```

Q.2)Write a PHP Script to accept customer Name from user and do the following
a) Transform Customer Name all Upper case latter.
Make First character to Upper Case.

```
<!DOCTYPE html>
<html>
<head>
    <title>Customer Name Transformation</title>
</head>
<body>

<form method="post">
    Enter Customer Name:
    <input type="text" name="custname">
    <input type="submit" value="Submit">
</form>

<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
```

```
    $name = $_POST['custname'];

    echo "<h3>Results:</h3>";

    // a) Convert to Uppercase
    echo "Uppercase: " . strtoupper($name) . "<br>";

    // b) First character uppercase
    echo "First Letter Capital: " . ucfirst($name);
}
?>

</body>
</html>
```

Q.1) Write inline CSS program to display with background color pink with red colored text.

```
<!DOCTYPE html>
<html>
<head>
    <title>Inline CSS Example</title>
</head>
<body style="background-color:pink;">

    <h2 style="color:red;">This is a heading with red text</h2>

    <p style="color:red;">
        This page has a pink background and red colored text using inline CSS.
    </p>

</body>
</html>
```

Q.2) Write a menu driven program the following operation on an associative array
a) Reverse the order of each element's key-value pair. [Hint: array_flip()]
b) Traverse the element in an array in random order. [Hint: shuffle()]

```
<?php
```

```
// Declare associative array
$arr = array("a"=>10, "b"=>20, "c"=>30, "d"=>40);

echo "Menu:\n";
echo "1. Reverse key-value pairs\n";
echo "2. Traverse array in random order\n";

echo "Enter your choice: ";
$choice = trim(fgets(STDIN));

switch($choice) {

    case 1:
        echo "Original Array:\n";
        print_r($arr);

        // Reverse key-value pairs
        $flipped = array_flip($arr);

        echo "After Reversing Key-Value Pairs:\n";
        print_r($flipped);
        break;

    case 2:
        echo "Original Array:\n";
        print_r($arr);

        // Shuffle works on indexed arrays, so convert
        $temp = array_values($arr);
        shuffle($temp);

        echo "Random Order Traversal:\n";
        print_r($temp);
        break;

    default:
        echo "Invalid choice!";
}
?>
```

Q.1) Design a table which shows weekly time table of a specific class.

```
<!DOCTYPE html>
<html>
<head>
  <title>Weekly Time Table</title>
</head>
<body>

<h2 style="text-align:center;">FYBSc Weekly Time Table</h2>

<table border="1" cellspacing="0" cellpadding="10" align="center">

  <tr>
    <th>Day/Time</th>
    <th>10:00 - 11:00</th>
    <th>11:00 - 12:00</th>
    <th>12:00 - 01:00</th>
    <th>02:00 - 03:00</th>
  </tr>

  <tr>
    <th>Monday</th>
    <td>Math</td>
    <td>Physics</td>
    <td>Chemistry</td>
    <td>Lab</td>
  </tr>

  <tr>
    <th>Tuesday</th>
    <td>English</td>
    <td>Math</td>
    <td>Physics</td>
    <td>Lab</td>
  </tr>

  <tr>
    <th>Wednesday</th>
    <td>Chemistry</td>
    <td>English</td>
```

```
      <td>Math</td>
      <td>Sports</td>
    </tr>

    <tr>
      <th>Thursday</th>
      <td>Physics</td>
      <td>Chemistry</td>
      <td>English</td>
      <td>Lab</td>
    </tr>

    <tr>
      <th>Friday</th>
      <td>Math</td>
      <td>Physics</td>
      <td>Computer</td>
      <td>Library</td>
    </tr>

</table>

</body>
</html>
```

Q.2)Write a script to create XML file as 'Employee.xml'. The element of this xml file are as follows:

```
<Empdetails>
<Employee EMPno= Empname=>
<Salary>--------</Salary>
<Designation>-------</Designation>
</Employee>
</Empdetails>
```

```
<?php

// Create XML structure
$xml = new SimpleXMLElement("<Empdetails></Empdetails>");

// Add employee node with attributes
$employee = $xml->addChild("Employee");
```

```php
$employee->addAttribute("EMPno", "101");
$employee->addAttribute("Empname", "Rahul");

// Add child elements
$employee->addChild("Salary", "50000");
$employee->addChild("Designation", "Developer");

// Save XML file
$xml->asXML("Employee.xml");

echo "Employee.xml file created successfully!";

?>
```

Q.1) Write internal CSS program to display with background color black with white colored text.

```html
<!DOCTYPE html>
<html>
<head>
  <title>Internal CSS Example</title>

  <style>
    body {
      background-color: black;
      color: white;
    }

    h1 {
      color: white;
      text-align: center;
    }

    p {
      color: white;
      font-size: 18px;
    }
  </style>

</head>

<body>
```

```
    <h1>Welcome to My Page</h1>

    <p>
        This page uses internal CSS with black background and white text color.
    </p>

</body>
</html>
```

Q.2) Write a PHP script to accept Employee details (eno, ename, address) on first page.On second page accept earning (Basic, Da, HRA). On third page print Employee information(eno,ename, Address, BASIC, DA, HRA, TOTAL)

```
<?php
session_start();
?>

<!DOCTYPE html>
<html>
<head>
    <title>Employee Details</title>
</head>
<body>

<form method="post" action="emp2.php">

    Employee No: <input type="text" name="eno"><br><br>
    Employee Name: <input type="text" name="ename"><br><br>
    Address: <textarea name="address"></textarea><br><br>

    <input type="submit" value="Next">

</form>

</body>
</html>
```

Q.1 Write external CSS program to display with background color sky blue with blue colored text.

```
<!DOCTYPE html>
<html>
```

```
<head>
   <title>External CSS Example</title>
   <link rel="stylesheet" type="text/css" href="style.css">
</head>

<body>

   <h1>Welcome to External CSS Page</h1>

   <p>This page uses external CSS with sky blue background and blue text.</p>

</body>
</html>
body {
   background-color: skyblue;
   color: blue;
}

h1 {
   text-align: center;
   color: blue;
}

p {
   font-size: 18px;
}
```

Q.2)Write a Calculator class that can accept two values, then add them, subtract them,multiply them together, or divide them on request.

```
 <?php

class Calculator {
   public $a;
   public $b;

   // Constructor
   function __construct($x, $y) {
      $this->a = $x;
      $this->b = $y;
   }
```

```php
    // Addition
    function add() {
        return $this->a + $this->b;
    }

    // Subtraction
    function subtract() {
        return $this->a - $this->b;
    }

    // Multiplication
    function multiply() {
        return $this->a * $this->b;
    }

    // Division
    function divide() {
        if ($this->b != 0)
            return $this->a / $this->b;
        else
            return "Division by zero error!";
    }
}

// Example usage
$num1 = 20;
$num2 = 10;

// Create object
$calc = new Calculator($num1, $num2);

echo "Addition: " . $calc->add() . "<br>";
echo "Subtraction: " . $calc->subtract() . "<br>";
echo "Multiplication: " . $calc->multiply() . "<br>";
echo "Division: " . $calc->divide();

?>
```

Q.1) Write CSS using HTML which uses of text decoration, border, padding and margin.

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Properties Demo</title>

  <style>
    .box {
      border: 3px solid blue;       /* border */
      padding: 20px;               /* space inside border */
      margin: 30px;                /* space outside border */
      text-decoration: underline;   /* text decoration */
      font-size: 20px;
      color: darkblue;
    }

    .box2 {
      border: 2px dashed red;
      padding: 15px;
      margin: 20px;
      text-decoration: line-through;
      color: red;
    }
  </style>

</head>

<body>

  <div class="box">
    This is a text with underline, border, padding and margin.
  </div>

  <div class="box2">
    This is another text with strike-through and different styling.
  </div>

</body>
</html>
```

Q.2)Write a PHP script for the following. Design a form to accept a string and check whether the given string is Palindrome or not.

```
<!DOCTYPE html>
<html>
<head>
  <title>Palindrome Checker</title>
</head>
<body>

<h2>Check Palindrome String</h2>

<form method="post">
  Enter String:
  <input type="text" name="str">
  <input type="submit" value="Check">
</form>

<?php

if ($_SERVER["REQUEST_METHOD"] == "POST") {

  $str = $_POST['str'];

  // Remove spaces and convert to lowercase
  $cleanStr = strtolower(str_replace(" ", "", $str));

  // Reverse string
  $revStr = strrev($cleanStr);

  echo "<h3>Result:</h3>";

  if ($cleanStr == $revStr) {
    echo "$str is a Palindrome String";
  } else {
    echo "$str is NOT a Palindrome String";
  }
}

?>
```

```
</body>
</html>
```

Q.1 Write CSS using HTML which displays following output Unordered lists

o Coffee
o Tea
o Milk
▪ Apple
▪ Mango
▪ Banana
Ordered list
I. Rose
II. Jasmine
III. Marigold
a. Sunflower
b. Tulip
c. Lily

```
<!DOCTYPE html>
<html>
<head>
  <title>List Styling Using CSS</title>

  <style>
    body {
      font-family: Arial;
    }

    .ul1 {
      list-style-type: circle;   /* o style */
      color: brown;
    }

    .ul2 {
      list-style-type: square;   /* ▪ style */
      color: green;
    }

    .ol1 {
      list-style-type: upper-roman;  /* I, II, III */
      color: blue;
    }

    .ol2 {
      list-style-type: lower-alpha;  /* a, b, c */
      color: purple;
```

```
    }
  </style>

</head>

<body>

<h2>Unordered List</h2>

<ul class="ul1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

<ul class="ul2">
  <li>Apple</li>
  <li>Mango</li>
  <li>Banana</li>
</ul>

<h2>Ordered List</h2>

<ol class="ol1">
  <li>Rose</li>
  <li>Jasmine</li>
  <li>Marigold</li>
</ol>

<ol class="ol2">
  <li>Sunflower</li>
  <li>Tulip</li>
  <li>Lily</li>
</ol>

</body>
</html>
```

(15)

Q.2) Write a PHP script to print following floyd’s triangle.

1
2 3
4 5 6

```
<?php

$n = 3;   // number of rows
$num = 1;
```

```php
for ($i = 1; $i <= $n; $i++) {

   for ($j = 1; $j <= $i; $j++) {
      echo $num . " ";
      $num++;
   }

   echo "<br>";
}

?>
```

Q.1) Write the JavaScript to convert temperature from Celsius to Fahrenheit.

```html
<!DOCTYPE html>
<html>
<head>
   <title>Temperature Conversion</title>
</head>
<body>

<h2>Celsius to Fahrenheit Converter</h2>

Enter Celsius:
<input type="text" id="celsius">

<button onclick="convert()">Convert</button>

<p id="result"></p>

<script>
function convert() {
   let c = document.getElementById("celsius").value;

   // Convert to Fahrenheit
   let f = (c * 9/5) + 32;

   document.getElementById("result").innerHTML =
      "Fahrenheit: " + f;
}
</script>
```

</body>
</html>

Q.2)Write a PHP program to define Interface shape which has two method as area() and volume (). Define a constant PI. Create a class Cylinder implement this interface and calculate area and Volume.

```
<?php

// Define constant PI
define("PI", 3.14);

// Interface
interface Shape {
   public function area();
   public function volume();
}

// Cylinder class implementing interface
class Cylinder implements Shape {

   public $radius;
   public $height;

   function __construct($r, $h) {
      $this->radius = $r;
      $this->height = $h;
   }

   // Area = 2πr(r + h)
   public function area() {
      return 2 * PI * $this->radius * ($this->radius + $this->height);
   }

   // Volume = πr²h
   public function volume() {
      return PI * $this->radius * $this->radius * $this->height;
   }
}

// Create object
```

```
$c = new Cylinder(5, 10);

echo "Surface Area of Cylinder: " . $c->area() . "<br>";
echo "Volume of Cylinder: " . $c->volume();

?>
```